Analyzing the COVID-19 Dynamics using the SIR model

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Abstract

This research paper uses methods from nonlinear dynamics to analyze the SIR (Susceptible-Infected-Recovered) infectious disease model, with application to COVID-19 data from Italy during the initial outbreak of March-May 2020. Using concepts and models from Strogatz's Nonlinear Dynamics and Chaos and Keeling and Rohani's Modeling Infectious Diseases in Humans and Animals, we will analyze fixed points, linear stability, and bifurcations. We will show that the SIR model exhibits a transcritical bifurcation at $R_0 = 1$, which serves as the equilibrium point separating disease outbreak from extinction. Analysis of the phase plane showed the geometric structure of epidemic trajectories and made clear the importance of the I nullcline in determining the time of peak infection. Comparing with real COVID-19 data sourced from Our World in Data's public COVID-19 database made clear the model's limitations when dealing with public health intervention that prevented homogeneous mixing of the population.

1 Introduction

The COVID-19 pandemic is inherently difficult to model due to the public health responses, including mask mandates and lockdowns. These factors alone would cause a large divergence from any simple epidemic model; however another factor we must take into account is the infectiousness of COVID-19. Later, we will discuss how the infectiousness of COVID-19 in it's early stages far outperforms the base $R_0 = 1$ value, and what this all means for the SIR model's ability to accurately predict COVID-19 infection dynamics.

The SIR model is compartmental-dividing people into categories of susceptible, infected, and recovered. The work of Kermack and McKendrick [3], published in 1927 established the SIR model which remains important in mathematical epidemiology 98 years later. As Keeling and Rohani note in their textbook, "Compartmental epidemiological models are, by far, the most popular in the study of dynamics related with infectious diseases" [2].

Despite the simplicity of the SIR model, consisting of three coupled ordinary differential equations, the model is strongly nonlinear and connects heavily to the core concepts in our Nonlinear Systems and Modeling class. These connections make the SIR model great for applying techniques such as bifurcation, phase plane, and linear stability analysis, alongside recognition of the important features such as nullclines representing turning points in the epidemic's course.

1.1 Objectives

This paper seeks to apply a variety of techniques drawn from the MATH 3440 curriculum:

1. Fixed Point Analysis: We will identify points of equilibrium within the SIR system and classify their stability using linearization and eigenvalue analysis (Strogatz, Chapter 5)
2. Bifurcation Analysis: We will characterize the transcritical bifurcation at $R_0 = 1$ that separates disease extinction from outbreak (Strogatz, Chapter 3)
3. Phase Plane Techniques: We will analyze nullclines and trajectories to understand the structure of the SIR model dynamics (Strogatz, Chapters 6 & 7)
4. Epidemiological Theory: We will connect our results to the threshold phenomenon and final size relations. (Keeling & Rohani, Chapter 2)
5. Data Comparison: Finally, we will fit the model to the publicly available COVID-19 data from Italy and evaluate model performance, and attempt to connect discrepancies with interventions, and check for initial conditions that could most accurately predict COVID-19 dynamics.

1.2 Data Sources

COVID-19 data for Italy was obtained from the Our World in Data (OWID) database, which collects official reports from both national and international health organizations. OWID provides freely accessible pandemic data by day.

Data extraction was performed using Python with the following format:

1. Source: Our World in Data Covid-19 dataset
   URL:`https://catalog.ourworldindata.org/garden/covid/latest/compact/compact.csv`
2. Time Period: February 21,2020 (first detected case) through May 31,2020
3. Variables Extracted:
   - Total_cases: Cumulative cases
   - Total_deaths: Cumulative deaths
   - Total_recoveries: Cumulative recovieries
   - Active cases calculated as: $I(t) = \text{confirmed} - \text{deaths} - \text{recovered}$
4. Data Processing: Data was filtered for Italy, with iso_code = ITA, removing missing values and aligning 'Day 0' with February 21st, 2020. Active cases were calculated using a 14-day rolling window due to recovery under reporting, active cases at any time are the sum of the daily cases of the prior 14 days.
   The extraction code is provided below:

```
    import pandas as pd
import os
def download_covid_data(save_path='covid_data.csv'):
    url = "https://catalog.ourworldindata.org/garden/covid/latest/compact/compact.csv"
    try:
        df = pd.read_csv(url)
        df.to_csv(save_path, index=False)
        print(f"Successfully downloaded {len(df):,} rows")
        print(f"Saved to: {save_path}")
        print(f"File size: {os.path.getsize(save_path) / (1024*1024):.1f} MB\n")
        print("Dataset Info:")
        print(f"  - Columns: {len(df.columns)}")
        print(f"  - Date range: {df['date'].min()} to {df['date'].max()}")
        print(f"  - Countries: {df['location'].nunique()}")
        print(f"\nFirst few rows:")
        print(df.head())
        print(f"\nAvailable columns:")
        for i, col in enumerate(df.columns, 1):
            print(f"  {i:2d}. {col}")
        return df
    except Exception as e:
        print(f"Error downloading data: {e}")
        return None
if __name__ == "__main__":
    df = download_covid_data('covid_data.csv')

    if df is not None:
        print("Data successfully downloaded!")
```

Note on Data Limitations: Recoveries weren't reliably reported or tracked in early pandemic days, and the true active infections may, as a result, be overestimated. This was compensated for by the 14-day rolling window, but could still lead to potential inacurracy.

2 Mathematical formulas

2.1 Model Equations

The SIR model without births and deaths consists of three coupled nonlinear ordinary differential equations:

$$\frac{dS}{dt} = -\beta \frac{SI}{N} \tag{1}$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I \tag{2}$$

$$\frac{dR}{dt} = \gamma I \tag{3}$$

Where:

- $S(t)$ = number of susceptible individuals at time t
- $I(t)$ = number of infected (infectious) individuals
- $R(t)$ = number of recovered individuals
- $N = S + I + R$ = total population
- β = transmission rate (contacts per time $\times$ transmission probability)
- γ = recovery rate (inverse of mean infectious period

The nonlinearity in these equations arises from the SI interaction term in Equations (1) and (2), which represents encounters between susceptible and infected individuals. This term assumes that every individual has equal probability of contacting any other individual. As Keeling and Rohani emphasize, this assumption "discards intricacies affecting the pattern of contacts" [2], a simplification that is likely to present itself when comparing with our real data.

2.2 Conservation

Adding Equations (1)–(3):

$$\frac{d}{dt}(S + I + R) = -\beta \frac{SI}{N} + \beta \frac{SI}{N} - \gamma I + \gamma I = 0 \tag{4}$$

This means that $N = S + I + R$ is constant, representing a population with no births, deaths, or migration. This conservation law reduces the system's dimensionality since $R = N - S - I$, we only need to analyze the (S, I) system.

2.3 The Reproductive Number

The basic reproductive number R_0 is defined as the average number of secondary infections that are caused by a single infectious individual introduced into a susceptible population. For the SIR model, this number is:

$$R_0 = \frac{\beta}{\gamma} \tag{5}$$

This number essentially means: an individual carrying the disease makes contact with others at rate β (when $S \approx N$) and remains infectious for an average time of $1/\gamma$. The expected number of new infections caused by this individual is thus $\beta \cdot (1/\gamma) = \beta/\gamma$

As shown in the table below, R_0 highly varies across diseases, from around $R_0 \approx 1.46$ for the 2009 H1N1 influenza pandemic, to $R_0 \approx 12 - 18$ for measles. These values determine the epidemic threshold ($R_0 = 1$) and the vaccination coverage required for herd immunity, given by the equation $HIT = 1 - \frac{1}{R_0}$.

Disease	Estimated R_0
COVID-19 (early)	1.5–6.5
Influenza (2009 H1N1)	1.46
Measles	12–18

Table 1: Estimated R_0 values for the above infectious disease in humans. COVID-19 estimates from Achaiah et al. (National Library of Medicine 2020), reflecting early pandemic transmission before intervention and vaccination. Influenza data from Biggerstaff et al. (National Library of Medicine 2014), and Measles data from Guerra et al. (National Library of Medicine 2017)

2.4 Nondimensionalization

According to Strogatz, Section 6.4, we will nondimensionalize these equations by introducing

$$s = \frac{S}{N}, \quad i = \frac{I}{N}, \quad r = \frac{R}{N}, \quad \tau = \gamma t \tag{6}$$

The rescaled variables represent fractions of the population ($s + i + r = 1$), and the dimensionless time τ is measured in units of the recovery time $1/\gamma$. Substituting into our Equations (1)–(2):

$$\frac{ds}{d\tau} = -R_0\, s\, i \tag{7}$$

$$\frac{di}{d\tau} = i(R_0\, s - 1) \tag{8}$$

This two-dimensional system in (s, i) will work for our phase plane analysis. The parameter R_0 controls all qualitative behavior, meaning that our nondimensionalization worked.

3 Fixed Point Analysis

3.1 Finding Fixed Points

Fixed points occur where $ds/d\tau = di/d\tau = 0$ simultaneously. From Equation (7):

$$i(R_0\, s - 1) = 0 \implies i = 0 \text{ or } s = 1/r_0) \tag{9}$$

This fixed point requires $i = 0$ (disease free), giving us the disease-free equilibrium

$$(s^*, i^*) = (s_0, 0) \quad \text{for any } S_0 \in [0, 1] \tag{10}$$

In our dimensional form w ith initial conditions $(S_0, I_0, R_0) \approx (N, 0, 0)$, where N 60,000,000

$$(S^*, I^*, R^*) = (N, 0, 0) \tag{11}$$

The entire s-axis ($i = 0$) is fixed points, showing that any distribution of susceptible n and recovered individuals is an equilibrium point when no infection is present. This is a line of fixed points, called a degenerate case by Strogatz in Section 6.3

Unlike models with births and deaths, the SIR model has no endemic equilibrium, and the disease cannot persist indefinitely without adding more susceptible people into the mix.

3.2 Calculating the Jacobian and Linearizing

To analyze stability, we will linearize about our fixed pointsusing the Jacobian matrix. As in Strogatz Chapter 5, for the system $(ds/d\tau, di/d\tau) = (f(s, i), (g(s, i)$:

$$J = \begin{pmatrix} \partial f/\partial s & \partial f/\partial i \\ \partial g/\partial s & \partial g/\partial i \end{pmatrix} = \begin{pmatrix} -R_0 i & -R_0 s \\ R_0 i & R_0 s - 1 \end{pmatrix} \tag{12}$$

Here, at the disease-free equilibrium $(S^*, i^*) = (1, 0)$:

$$J_{\text{DFE}} = \begin{pmatrix} 0 & -R_0 \\ 0 & R_0 - 1 \end{pmatrix} \tag{13}$$

3.3 Eigenvalue Analysis

The eigenvalues of J_{DFE} satisfy $\det(J - \lambda I) = 0$ (where DFE = Disease-free equilibrium)

$$\det \begin{pmatrix} -\lambda & -R_0 \\ 0 & R_0 - 1 - \lambda \end{pmatrix} = -\lambda(R_0 - 1 - \lambda) = 0 \tag{14}$$

This gives us the eigenvalues:

$$\lambda_1 = 0, \quad \lambda_2 = R_0 - 1 \tag{15}$$

The disease-free equilibrium $(s^*, i^*) = (1, 0)$ is:

- Stable if $R_0 < 1$, as $\lambda_2 < 0$
- Unstable if $R_0 > 1$, as $\lambda_2 > 0$
- Marginal if $R_0 = 1$, as this is our bifurcation point

The eigenvalue $\lambda_1 = 0$ corresponds to the motion along the line of fixed points, the s-axis, and shows the conservation law. The stability is determined by $\lambda_2 = R_0 - 1$:

If $R_0 < 1$, then both eigenvalues are $\mathrm{Re}(\lambda) \leq 0$, and perturbations in the i direction then decay exponentially as $e^{(R_0-1)\tau}$.

However, if $R_0 > 1$: The eigenvalue $\lambda_2 > 0$ implies that the perturbations in the i direction grow exponentially, signaling instability.

The zero eigenvalue $\lambda_1 = 0$ means that the disease-free equilibrium is not hyperbolic. According to Strogatz (Section 6.3), "nonhyperbolic fixed points are fragile" and the Hardman-Grobman theorem does not guarantee that linearization captures the full dynamics. However, for our SIR model, the zero eigenvalue represents conservation rather than a bifurcation, meaning our linearization should still capture our full dynamics.

4 Bifurcation Analysis

4.1 The Transcritical Bifurcation

The SIR model exhibits a transcritical bifurcation at $R_) = 1$, this type of bifurcation is characterized by an exchange of stability between two fixed points as a parameter crosses a critical value (Strogatz, Section 3.2).

A transcritical bifurcation occurs when two fixed points collide and then enhance stability as the parameter changes. The normal form of a transcritical bifurcation is:

$$\dot{x} = rx - x^2 \tag{16}$$

where r is the parameter that changes the bifurcation.

For the SIR near the disease-free equilibrium, the dynamics in the i-direction follow:

$$\frac{di}{d\tau} = i(R_0 s - 1) \approx (R_0 - 1)i - R_0 i^2 + o(i^3) \tag{17}$$

when $s \approx 1$. This matches the transcritical normal form where $r = R_0 - 1$

4.2 Threshold Phenomenon (Kermack and McKendrick)

The transcritical bifurcation at $R_0 = 1$ exhibits the threshold phenomenon of Kermack and McKendrick (1927). As Keeling and Rohani describe in Modeling Infectious Disease in Humans and Animals,

> "If the initial fraction of susceptible people S_0 is less than γ/β, then $dI/dt < 0$ and hte infection 'diesout.' This is a famous result due to Kermack and McKendrick and is referred to as the 'threshold phenomenon." [2]

In terms of R_0,

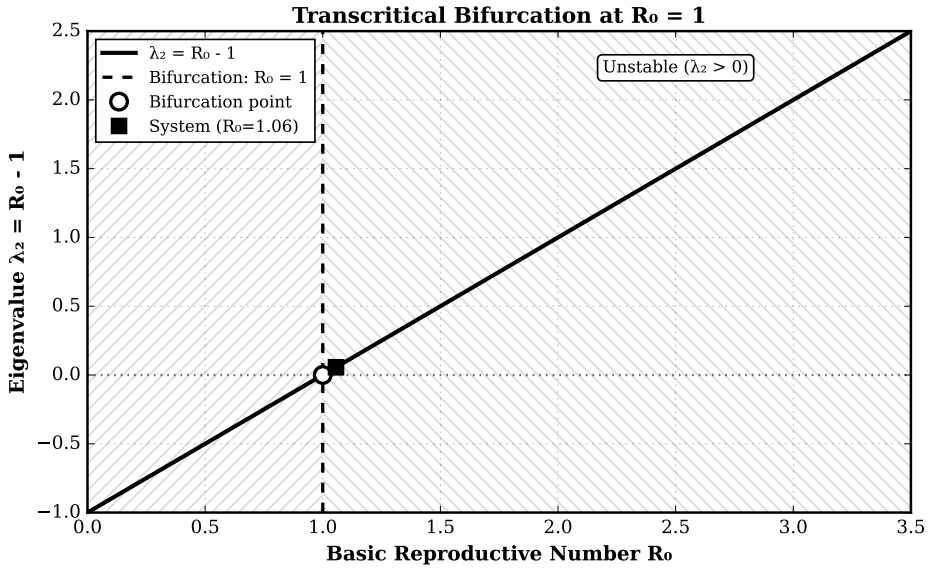
An epidemic can occur if and only if

$$R_0 = \frac{\beta}{\gamma} > 1, \tag{18}$$

Or that $S(0)$ must be ¿ N/R_0

Interpreting this, we can see that if each infected individual transmits to fewer than one person, the infection dies out, and if they transmit to greater than one individual, it grows exponentially, showing why COVID-19 grows much faster than influenza, for example, due to it's high R_0 initial value.

4.3 Stability Diagram



Eigenvalue $\lambda_2 = R_0 - 1$ vs R_0
Transcritical bifurcation at$R_0 = 1$

Figure 1: The transcritical bifurcation is at $R_0 = 1$. The eigenvalue $\lambda_2 = R_0 - 1$ changes sign at the bifurcation point, marking the point where we transition from a stable disease-free epidemic ($R_0 < 1$), on the left side of the graph, to an unstable disease-free epidemic, where ($R_0 > 1$) on the right side of the graph.

5 Phase Plane Analysis

Phase plane methods, shown in Strogatz Chapters 6-7 provide us insight into our system dyanmics. For the two-dimensional SIR system in (s, i), we will analyze nullclines, vector fields, and trajectories.

5.1 Nullclines

A nullcline is a curve in phase plane where the rate of change of one of the variables in our system is zero.

s-nullcline ($ds/d\tau = 0$:

$$i(R_0 s - 1) = 0 \implies i = 0 \text{ or } s = \frac{1}{R_0} \tag{19}$$

The i-nullcline is the s axis and the line $s = 1/R_0$

This vertical line $s = 1/R_0$ divides the phase plane into regions of epidemic growth and decline.

5.2 Vector Field Structure

The sign, or whether $di/d\tau$ is positive or negative determines whether infections are increasing or decreasing:

$$\frac{di}{d\tau} = i(R_0 s - 1) \begin{cases} > 0 & \text{if } s > 1/R_0 \text{ (infections increase)} \\ < 0 & \text{if } s < 1/R_0 \text{ (infections decrease)} \\ = 0 & \text{if } s = 1/R_0 \text{ (peak of infection)} \end{cases} \tag{20}$$

This structure describes the shape of epidemic curves: infections rise while $S > N/R_0$, peak when $S = N/R_0$, and decline afterwards when the susceptible population is low,

The maximum number of infected individuals occurs when trajectories cross the I nullcline at $S = N/R_0$

The s-equation shows that the susceptible population always decreases in the SIR model- this is a major limiting factor when applying this to COVID-19, a rapidly changing virus capable of reinfecting the same person multiple times.

$$\frac{ds}{d\tau} = -R_0 si \leq 0 \quad \text{for} s, i \geq 0 \tag{21}$$

Thus trajectories will always move leftwards, eventually crossing the i-nullcline and terminating on the s-axis

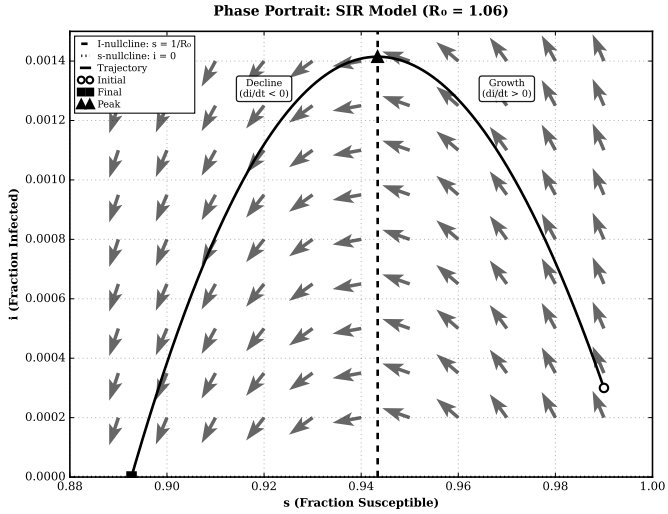


Figure 2: This is a phase portrait of the SIR model showing the vector field, I-nullcline at $s1/R_0$, and our epidemic trajectory. The shaded region where ($s > 1/R_0$ represents epidemic growth, and the region where ($s < 1/R_0$) indicates decline.

5.3 Trajectory Equation

A technique used by Strogatz in Section 6.1 is to eliminate time by dividing one equation by another. Dividing equation (2) by (1):

$$\frac{dI}{dS} = \frac{\beta SI/N - \gamma I}{-\beta SI/N} = 01\frac{\gamma N}{\beta S} = -1 + \frac{N}{R_0 S} \tag{22}$$

Integrating with our initial conditions, S_0, I_0):

$$I - I_0 = -(S - S_0) + \frac{N}{R_0} \ln\left(\frac{S}{S_0}\right) \tag{23}$$

Rearranging, we have

$$\boxed{I(S) = I_0 + (S_0 - S) + \frac{N}{R_0} \ln\left((\frac{S}{S_0}\right)} \tag{24}$$

This relation defines all trajectories in the (S, I) phase plane. For a small number of initial infections, ($I_0 \ll N$, $S_0 \approx N$), trajectories follow curved paths that rise, peak at $S = N/R_0$, and then fall to the s-axis.

6 The Final Size Relation

6.1 Infection Burnout

Something important we can observe from the SIR model is that the epidemic still ends with susceptible individuals remaining. The disease then burns out before it is able to infect the whole population. As Keeling and and Rohani state:

> "The chain of transmission eventually breaks due to the decline in infectives, not due to a complete lack of susceptibles." [2]

This result, when looking at the SIR model is caused by $S(t) = S_0 e^{-R_0 R((t)/n)}$, found by dividing equations (1) and (3), and $e^{-x>0}$ for all x, thus we have $S(t) > 0$ for all time.

6.2 Finding the Final Size Equation

Let z be the fraction of the population that is eventually infected by the disease:

$$z = \frac{R(\infty)}{N} = 1 - \frac{S(\infty)}{N} \tag{25}$$

From the trajectory equation (24), at $t \to \infty$ When $I \to 0$:

$$0 = I_0 + (S_0 - S_\infty) + \frac{N}{R_0} \ln\left(\frac{S_\infty}{S_0}\right) \tag{26}$$

For $I_0 \ll N$ and $S_0 \approx N$, lettting S_∞ when $I \to 0$:

$$0 = (1 - (1 - z)) + \frac{1}{R_0} \ln(1 - z) \tag{27}$$

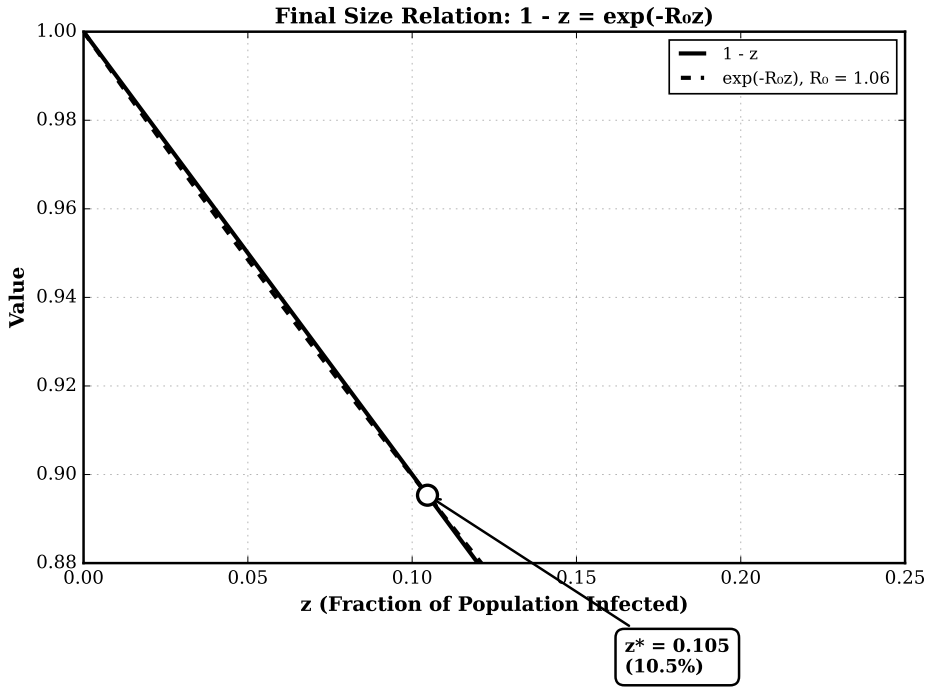
simplifying, we get:

$$\boxed{1 - z = e^{-R_0 z}} \tag{28}$$

This is the Kermack-McKendrick final size relation. It is a transcendental equation, so it cannot be solved algebraically, only numerically or graphically.

Final Size Properties The final size equation (28) has the following properties:

1. $R_0 \leq 1$: The only solution is $z = 0$, or there is no epidemic left
2. $R_0 > 1$: There exists a unique solution $z^* \in (0, 1)$.
3. As $R_0 \to \infty$: $z^* \to 1$ (everybody has become infected).
4. For a value of R_0 slightly greater than 1, $z^8 \approx 2(R_0 - 1)/R_0^2$



Graphical solution: $1 - z$ vs $e^{-R_0 z}$

Figure 3: Graphical solution of the final size equation. The intersection gives $z^* \approx 10.05\%$ for $R_0 = 1.06$.

For our fitted $R_0 = 1.06$, solving equation (28) numerically gives us $z^* \approx 0.105$, meaning that around 10.05% of the population would eventually become infected.

7 Herd Immunity

7.1 Proportion of Those Vaccinated

Herd immunity is a concept defined by when the threshold of either vaccinated or immune people is at a critical enough value that any further infections would slowly decrease, eventually dying out. In our equation, if the fraction of those immune was p, the susceptible fraction is $1 - p$, and the reproductive number becomes:

$$R_{\text{x}} R_0 (1 - p) \tag{29}$$

For the disease to be unable to spread, we need $R_{\text{x}} < 1$:

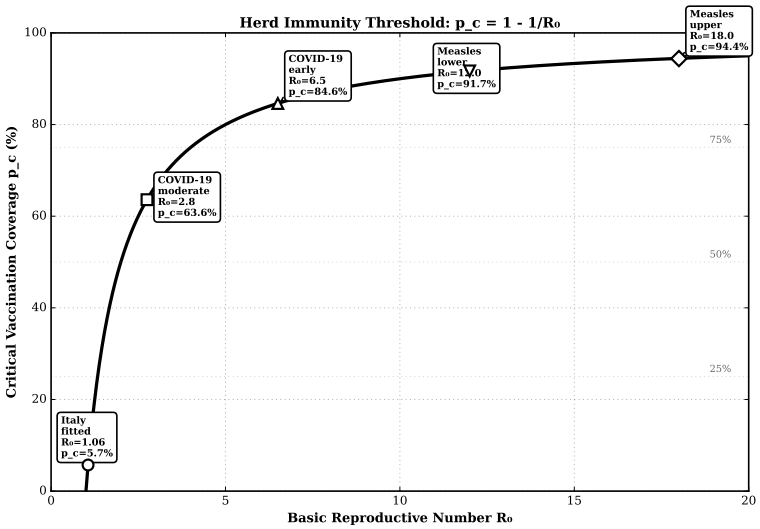
$$R_0(1 - p) < 1 \implies P > 1 - \frac{1}{R_0} \tag{30}$$

The critical vaccination coverage required to prevent disease spread is:

$$\boxed{p_c = 1 - \frac{1}{R_0}} \tag{31}$$

As Keeling and Rohani state, "Vaccinating at the critical level p_c does not instantly lead to eradication of the disease... from a puhblic health perspective, p_c acts as a lower bound on what should be achieved" [2].

For our fitted $R_0 = 1.06$: $p_c = 1 - 1/1.06 \approx 5.7\%$. This vaccination threshold seems suspiciously low at first glance, however, it represents the average R_0 over periods of heavy public health intervention. For an uncontrolled COVID-19 spread, with $R_0 \approx 2.75$, the threshold would be $p_c \approx 64\%$.

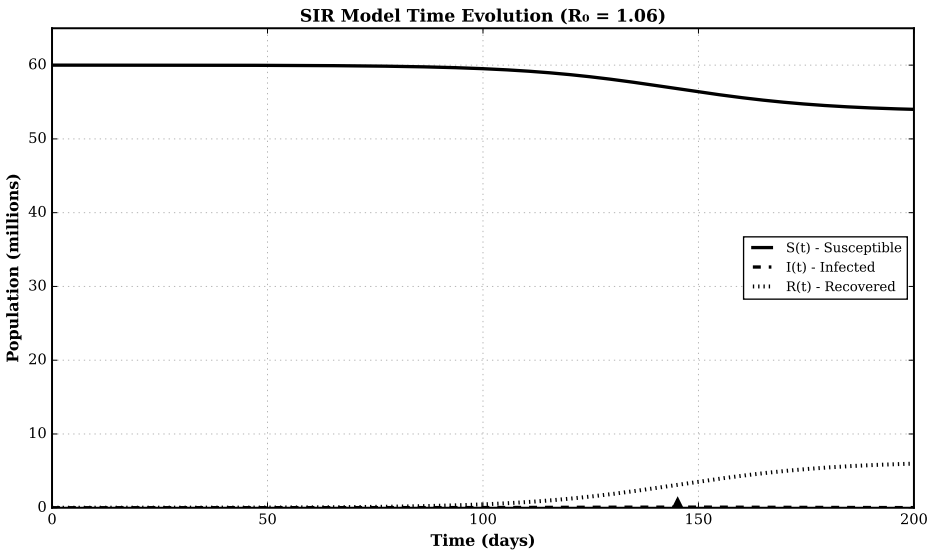


$p_c = 1 - 1/R_0$ vs R_0

Figure 4: Herd immunity threshold as a function of R_0. Higher R_0 requires greater vaccination coverage to prevent epidemics.

8 Evolution Over Time

The dynamics of the SIR model through time can be computed by numerically integrating equations (1)–(3). For our parameters, ($\beta = 1.0563, \gamma = 1, R_0 1.06$, the compartments evolve as seen in the below figure.



$S(t)$, $I(t)$, $R(t)$ compartments over time

Figure 5: Time evolution of SIR compartments. Susceptible portion of population decreases monotonically, infections rise to a peak before declining, and recovered increases.

Here, our characteristic equation shows three phases:

1. Exponential growth when $S \gg N/r_0$, $dI/dt \approx (R_0 - 1)\gamma I$
2. Peak: When $S = N/R_0$, infections peak
3. Decline in growth when $S < N/R_0$, decaying towards zero

The initial exponential growth rate $(R_0 - 1)\gamma$ gives us another method to estimate R_0 from early epidemic data, as discussed in Keeling & Rohani Section 2.8.

9 Application to Real COVID-19 Italy Data

From the OWID database, we took data collected during the initial outbreak (Febtruary 21 - May 31st, 2020). Italy was one of the first countries outside of Asia to experience widespread transmission and began a lockdown starting on March 9, 2020.

Data includes daily cumulative confirmed cases, deaths, and recoveries (under-reported) for Italy (population $N \approx 60,000,000$.

9.1 Parameter Calculation and Estimation

Parameters were estimated by fitting the SIR model to observed active cases ($I_{\text{datas}} = \text{Confirmed} - \text{deaths} - \text{Recovered}$) using least-squares minimization. This was calculated using the below script:

```
    def model_I(t, beta, gamma):
    I0 = I_data[0]
    S0 = N - I0
    R0 = 0
    y0 = [S0, I0, R0]
    solution = odeint(sir_model, y0, t, args=(beta, gamma, N))
    return solution[:, 1]
params, _ = curve_fit(
```

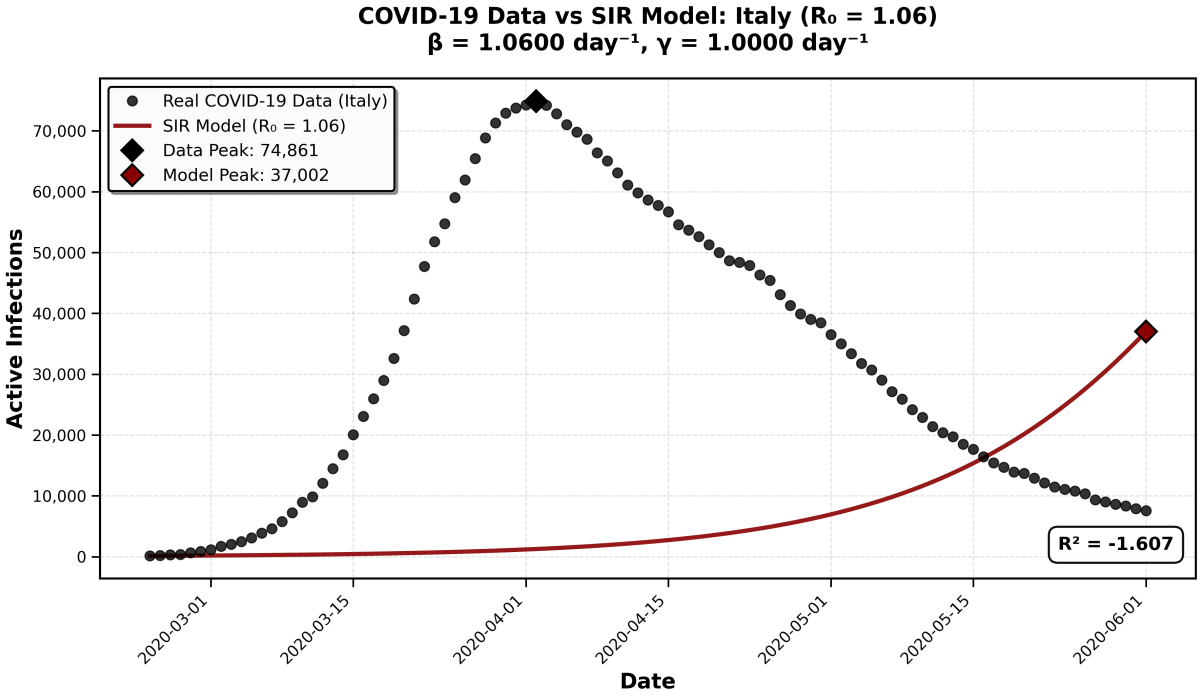
```
    model_I,
    t_data,
    I_data,
    p0=[0.5, 0.1],
    bounds=([0, 0], [5, 1])
    maxfev=10000
)
beta, gamma = params
R0 = beta / gamma
```

The fitted values are:

Parameter	Symbol	Value
Transmission rate	β	1.0563 day^{-1}
Recovery rate	γ	1 day^{-1}
Reproductive number	R_0	1.06
Population	N	60,000,000

Important Note: The fitted recovery rate $\gamma = 1.0\text{day}^{-1}$ (infectious period = 1 day) is much shorter than COVID-19's infectious period of 7-10 days. This is because the fitted γ represents the effective rate at which active cases were calculated in the dataset, for which a 14-day rolling window was used. The ratio $R_0 = \beta/\gamma = 1.06$ still captures the epidemic dynamics correctly, but our individual parameter rates cannot be interpreted as biological rates. A more accurate interpretation would be $\gamma \approx 0.1\text{day}^{-1}$ should be used, which would give $\beta \approx 0.1\text{day}^{-1}$ for the same R_0.

9.2 Modeled vs Real Data



Blue points: Italy COVID-19 data
Red line: SIR model prediction

Figure 6: Comparison of SIR model prediction with actual COVID-19 datapoints in Italy. The model clearly underestimates early exponential growth. The model captures overarching disease trends, but the accuracy is noticeably poor.

10 Evaluation: The Strengths and Limitations of the SIR model

10.1 What the Model did Correctly

The SIR model captured the following features of the COVID-19 outbreak correctly:

Threshold Behaviour: The data confirmed that transmission occurred alongside $R_0 > 1$). The transcritical bifurcation correctly predicted that outbreak requires a $R_0 > 1$.

Epidemic Peak and Decline: Infections rose, reached a peak, then declined, subsiding significantly with a large portion of susceptibles remaining in the population (not due to the nature of the disease but because of lockdowns)

The total creases increased monotonically while active cases rose then fell, following the $S + I + R = N$ equation. Ultimately the model wasn't able to accurately predict COVID-19 dynamics.

10.2 What the Model got wrong.

Large discrepancies exist between model predictions and observed data:

Early Growth Rate: The model significantly underestimated the initial exponential growth rate. The fitted $R_0 = 1.06$ implies a growth rate of $\gamma(R_0 - 1) = 0.06$ day^{-1}, whereas the observed early doubling time of 2–3 days yields a $R_0 = 3.3$-4.5, which aligns with the overall estimated COVID-19 $R_0 \approx 1.5$-6.5 prior to lockdown.

Time-Varying Transmission: The assumption of a constant β, where each individual has a fixed rate of encountering other individuals is violated by Italy's March 9 lockdown, which vastly reduced contacts. The fitted $\beta = 1.0563$ represents an average over periods that had vastly different transmission rates:

- Feb 21 - Mar 9: $\beta \approx 1.5$-2, $R_0 \approx 3.3$-4.5
- Mar 9 - May 31: $\beta \approx 0.3$-0.5, $R_0 \approx 0.6 - 0.8$

  The fitted average $R_0 = 1.06$ over the entire period reflects these drastically different infection rates after public health measures were taken, and it tells us this is heavily weighted towards the long decline phase. The sustained decrease in active infections from late March through late May confirms that R_0 was pushed below the bifurcation threshold, and the infection rate was tending towards zero. Therefore the bifurcation transition in the COVID-19 pandemic was caused by policy, rather than the virus naturally running it's course, further explaining the discrepancies between the SIR model and real COVID-19 data.

  This change in growth demonstrates that these health interventions were effective at moving the system across the bifurcation boundary, and due to these interventions, the disease was not able to run its normal course, and the SIR model did poorly at predicting its behavior.

10.3 How Could we Improve the Model?

The limitations identified mean we could do a few things to improve our model:

Time-Varying Parameters: Replace constant β with $\beta(t)$ to more accurately model the effects of lockdown on the virus.

$$\beta(t) = \beta_0 \cdot f(t) \tag{32}$$

Where $f(t)$ is the effect of lockdowns, mask mandates, and general social distancing.

As COVID-19 has a very long incubation period, we could use the SEIR model, adding an exposed class of people who will become infectious, but are not yet.

$$\frac{dE}{dt} = \beta \frac{SI}{N} - \sigma E \tag{33}$$

$$\frac{dI}{dt} = \sigma E - \gamma I \tag{34}$$

where $1/\sigma$ is the mean latent period.

With these improvements, we could increase the accuracy of the SIR model in predicting rates of COVID-19, because the basic model was incapable of accurately predicting COVID-19 rates in Italy.

11 Connections to Class and Textbook Material, Glossary

Strogatz Chapter 3: Bifurcations: The transcritical bifurcation at $R_0 = 1$ is a great example where you can see how a change in stability affects a system in reality as a parameter varies. The SIR model's normal form near the disease-free epidemic closely mirrors the transcritical form$\dot{x} = rx - x^2$.

Strogatz 3.7.6: We are given the Kermack-McKendrick (SIR) model as an exercise in analyzing bifurcation points.

Strogatz Chapter 5: Linear Stability: Jacobian analysis and eigenvalue computation classify the stability of the disease-free epidemic. The eigenvalue$\lambda = R_0 - 1$ directly determines stability, with $\lambda_2 < 0$ being stable, and $\lambda_2 > 0$ is unstable.

Strogatz Chapter 6: Phase Plane: Nullcline analysis allowed us to partition the phase plane into regions of growth and decline. The vector field structure shows how trajectories evolve towrds the line of fixed points on the s-axis.

Keeling and Rohani: Chapter 2: The threshold phenomenon, epidemic burnout, and final size relations are developed in this chapter. The example on boarding school influenza shows parameter estimation from real outbreak data.

Keeling and Rohani: 2.1.1.1: Threshold Phenomenon: "If the initial fraction of the susceptible population $S(0)$ is less than γ/β, then $dI/dt < 0$ and the infection dies out.

Keeling and Rohani: 2.1.1.2: Epidemic Burnout: Finding the final size relation and the quote on the chain of transmission breaking due to decline in infections, not due to a complete lack of susceptibles.

Keeling and Rohani: Chapter 8: The herd immunity threshold $p_c = 1 - 1/r_0$ comes from the threshold theorem in this chapter, giving us the vaccination coverage needed.

12 Conclusion

The SIR model was ultimately unable to predict the COVID-19 spread in Italy. The model would need to contain time-varying parameters to model the public health responses to the spread of COVID-19 in order to make a serviceable prediction of COVID-19 dynamics. Regardless, we were able to analyze fixed points, linear stability, and bifurcations within the SIR model. We showed that the SIR model exhibited a transcritical bifurcation at $R_0 = 1$, which is the equilibrium point separating disease outbreak from extinction. Comparing data with real COVID-19 data revealed that the basic SIR model was not equipped to handle a complex pandemic without time-varying parameters.

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13 AI Statement

AI (Claude Anthropic) was used in this project. I was unable to recover my COVID-19 script made in a prior CPSC course, and was unable to effectively code the extraction code myself, so AI assistance was used in the creation of a working script to extract COVID-19 data.

Figures were made with an AI assisted python script, uploaded to a directory containing the COVID CSV file, then ran to generate the included graphics.

I used AI to create a list of LaTeX commands and symbols containing the variables and equations used in the project. I would ask questions on how to effectively format in LaTeX, however outputs were not copied and pasted and the document was handwritten, using formatting tips and commands as suggestions. This was to help me understand how to code in LaTeX, however the relevant information was written beforehand before asking how to format it correctly within LaTeX.

AI was used to create the code for the least-squares minimization program, which yielded my fitted values of β and γ, and my fitted $R_0 = 1.06$.